

Formulas and Tips

A quick-reference cheat sheet, grouped by topic:

- 1. Number & Arithmetic
- 2. Algebra
- 3. Geometry & Measurement
- 4. Counting & Probability
- 5. Reasoning Aids (Verbal & Numerical)

1. Number & Arithmetic

Number, Average, Percentage & Rate

Description	Formula
HCF and LCM relationship	$HCF(a, b) \times LCM(a, b) = a \times b$
Mean (average)	$\frac{\text{sum of values}}{\text{number of values}}$
Average speed	$\frac{\text{total distance}}{\text{total time}}$ (not the average of the speeds)
Speed / Distance / Time	$\text{speed} = \frac{\text{distance}}{\text{time}}$
Percentage change	$\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$
Simple interest	$I = \frac{P \cdot R \cdot T}{100}$
Profit / Loss percentage	$\frac{\text{profit or loss}}{\text{cost price}} \times 100\%$

Divisibility Quick Checks

- **2:** last digit even. **5:** ends in 0 or 5. **10:** ends in 0.
- **4:** last two digits form a number divisible by 4. **8:** last three digits divisible by 8.
- **3:** digit sum divisible by 3. **9:** digit sum divisible by 9. **6:** divisible by both 2 and 3.
- **11:** the alternating digit sum is 0 or a multiple of 11. e.g. 2728: $2 - 7 + 2 - 8 = -11 \checkmark$ (divisible); 2918: $2 - 9 + 1 - 8 = -14 \times$ (not divisible).

Index (Power) Laws

Law	Rule
Multiplying	$a^m \cdot a^n = a^{m+n}$
Dividing	$\frac{a^m}{a^n} = a^{m-n}$
Power of a power	$(a^m)^n = a^{m \cdot n}$
Power of a product	$(a \cdot b)^n = a^n \cdot b^n$
Zero index	$a^0 = 1$ (for $a \neq 0$)
Negative index	$a^{-n} = \frac{1}{a^n}$
Fractional index	$a^{\frac{1}{n}} = \sqrt[n]{a}$, and $a^{\frac{m}{n}} = \sqrt[n]{a^m}$

Sequences & Series

Type	nth term	Sum of first n terms
Arithmetic (common difference d)	$a + (n - 1)d$	$\frac{n}{2}(2a + (n - 1)d)$, or $\frac{n}{2}(\text{first} + \text{last})$
Geometric (common ratio r)	$a \cdot r^{n-1}$	$\frac{a(r^n - 1)}{r - 1}$ (for $r \neq 1$)

Useful sums (first n terms)

Description	Formula
Sum of the first n natural numbers	$\frac{n \cdot (n+1)}{2}$
Sum of the first n even numbers	$n \cdot (n + 1)$
Sum of the first n odd numbers	n^2
Sum of the first n squares	$\frac{n \cdot (n+1) \cdot (2 \cdot n+1)}{6}$
Sum of the first n cubes	$\left(\frac{n \cdot (n+1)}{2}\right)^2$

Number patterns to recognise on sight: - **Square numbers:** 1, 4, 9, 16, 25, . . . (n^2) - **Triangular numbers:** 1, 3, 6, 10, 15, . . . $\left(\frac{n(n+1)}{2}\right)$ - **Fibonacci:** 1, 1, 2, 3, 5, 8, 13, . . . (each term is the sum of the previous two) - **Powers of 2:** 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024

2. Algebra

Parabola (Quadratic) Formulas

For a quadratic $y = ax^2 + bx + c$ (with $a \neq 0$), the graph is a parabola — opening **upwards** if $a > 0$ and **downwards** if $a < 0$.

The three forms (each one tells you something at a glance)

Form	Equation	What you can read off immediately
Standard (general)	$y = ax^2 + bx + c$	y -intercept = $c \rightarrow$ point $(0, c)$
Vertex (turning point)	$y = a(x - h)^2 + k$	vertex (h, k) and axis of symmetry $x = h$
Factored (x -intercept)	$y = a(x - p)(x - q)$	x -intercepts at $x = p$ and $x = q$

Key formulas (standard form)

Description	Formula
Axis of symmetry	$x = -\frac{b}{2a}$
Vertex from standard form	$\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right)$
y -intercept	$(0, c)$

Finding the x -intercepts (roots)

Set $y = 0$ and solve $ax^2 + bx + c = 0$. Two ways: - **Factorise** if it factors neatly, e.g.

$x^2 - 5x + 6 = (x - 2)(x - 3) = 0 \Rightarrow x = 2, 3$. - **Read straight off the factored form**

$y = a(x - p)(x - q) \rightarrow$ roots are $x = p$ and $x = q$ (note the sign flips: $(x + 3)$ gives a root at $x = -3$).

The **axis of symmetry** is always halfway between the two x -intercepts: $x = \frac{p+q}{2}$.

Effect of the dilation factor a

a sets the **direction and width**: $a > 0$ opens **up** (vertex is a minimum), $a < 0$ opens **down** (vertex is a maximum). The **bigger** $|a|$ is, the **narrower** (steeper) the parabola; the **smaller** $|a|$ is (closer to 0), the **wider** (flatter) it is. At $a = \pm 1$ it has the same width as $y = \pm x^2$.

Notes

- The **axis of symmetry** passes through the vertex; the parabola is a mirror image on either side of it.
- The x -coordinate of the vertex is always $-\frac{b}{2a}$; substitute it back to get the y -coordinate.
- In vertex form $y = a(x - h)^2 + k$, the vertex is (h, k) , so the sign **inside the bracket** is opposite to the shift: $(x - 3)^2$ shifts **right** 3, and $(x + 3)^2$ shifts **left** 3. The k is straightforward: $+k$ up, $-k$ down.
- Finding a from a graph**: given the vertex (h, k) and one other point (x, y) , $a = \frac{y-k}{(x-h)^2}$.

Coordinate Geometry

For points (x_1, y_1) and (x_2, y_2) :

Description	Formula
Distance between two points	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
Gradient (slope)	$m = \frac{y_2-y_1}{x_2-x_1}$
Equation of a straight line	$y = mx + c$ (m = gradient, c = y -intercept)

- **Parallel** lines have **equal** gradients; for **non-vertical perpendicular** lines the gradients multiply to -1 . (A horizontal line and a vertical line are also perpendicular.)

3. Geometry & Measurement

Angle & Line Facts

Fact	Rule
Angles on a straight line	add to 180°
Angles around a point	add to 360°
Angles in a triangle	add to 180°
Angles in a quadrilateral	add to 360°
Vertically opposite angles	are equal
Exterior angle of a triangle	equals the sum of the two opposite interior angles

Parallel lines cut by a transversal: - **Corresponding** angles (F-shape) are **equal**. - **Alternate** angles (Z-shape) are **equal**. - **Co-interior / allied** angles (C or U-shape) add to 180° .

Polygon angles (n sides): - Sum of the **interior** angles = $(n - 2) \cdot 180^\circ$. - Sum of the **exterior** angles = 360° (for any polygon). - For a **regular** polygon: each interior angle = $\frac{(n-2) \cdot 180}{n}$; each exterior angle = $\frac{360}{n}$.

Area & Perimeter Formulas

Shape	Attributes	Area	Perimeter
Rectangle	length l and width w	$l \cdot w$	$2 \cdot (l + w)$
Square	side a	a^2	$4 \cdot a$
Parallelogram	base b , height h and slant side s	$b \cdot h$	$2 \cdot (b + s)$
Rhombus	side a , diagonals d_1 and d_2	$\frac{d_1 \cdot d_2}{2}$	$4 \cdot a$
Trapezium (Trapezoid)	bases b_1 , b_2 and height h	$\frac{(b_1 + b_2) \cdot h}{2}$	sum of all sides
Triangle	base b and height h	$\frac{b \cdot h}{2}$	sum of all sides
Equilateral Triangle	side a	$\frac{\sqrt{3} \cdot a^2}{4}$	$3 \cdot a$
Circle	radius r	$\pi \cdot r^2$	$2 \cdot \pi \cdot r$

Circle: Arc & Sector (angle θ in degrees)

Description	Formula
Diameter	$d = 2 \cdot r$
Arc length	$\frac{\theta}{360} \times 2\pi r$
Sector area	$\frac{\theta}{360} \times \pi r^2$

Volume & Surface Area Formulas

Shape	Attributes	Volume	Surface Area
Cube	side a	a^3	$6 \cdot a^2$
Rectangular Prism	length l , width w and height h	$(l \cdot w \cdot h)$	$(2 \cdot (l \cdot w + w \cdot h + h \cdot l))$
Cylinder	radius r and height h	$(\pi \cdot r^2 \cdot h)$	$(2 \cdot \pi \cdot r^2 + 2 \cdot \pi \cdot r \cdot h)$
Cone	radius r and height h	$\frac{\pi \cdot r^2 \cdot h}{3}$	$(\pi \cdot r^2 + \pi \cdot r \cdot \sqrt{r^2 + h^2})$
Sphere	radius r	$\frac{4 \cdot \pi \cdot r^3}{3}$	$4 \cdot \pi \cdot r^2$
Square-based Pyramid	base side a , height h , slant height l	$\frac{a^2 \cdot h}{3}$	$a^2 + 2 \cdot a \cdot l$

Notes

- For **any** pyramid or cone, Volume = $\frac{1}{3} \times$ base area $\times$ height (use the perpendicular height, not the slant height).
- The **slant height** of a square pyramid is $l = \sqrt{\left(\frac{a}{2}\right)^2 + h^2}$.
- Surface area = base + lateral area. For a square pyramid the 4 triangular faces give the lateral area $2 \cdot a \cdot l$.

Faces, Edges and Vertices of common 3D shapes

Shape	Faces	Edges	Vertices
Cube	6	12	8
Rectangular Prism	6	12	8
Triangular Prism	5	9	6
Square-based Pyramid	5	8	5
Triangular-based Pyramid (Tetrahedron)	4	6	4
Cylinder	3	2	0
Cone	2	1	1
Sphere	1	0	0

Euler's Formula (polyhedra with flat faces only)

$$V - E + F = 2$$

- Check (cube): $8 - 12 + 6 = 2 \checkmark$; (square pyramid): $5 - 8 + 5 = 2 \checkmark$; (triangular prism): $6 - 9 + 5 = 2 \checkmark$.
- It works **only** for solids with flat faces (prisms, pyramids) — **not** cylinders, cones, or spheres.

Note (school convention for curved solids)

- The counts above use the common school convention. Strictly (topologically), **curved surfaces have no edges or vertices**, so some exams count a cylinder and a cone as **0 edges / 0 vertices**. Read the question to see which version is expected.
- A **square-based pyramid** has 8 edges and 5 vertices; a **triangular-based pyramid (tetrahedron)** has 6 edges and 4 vertices — don't mix them up.

Trigonometric Formulas

Right Triangle

Description	Formula
Pythagorean Theorem	$a^2 + b^2 = c^2$
Sine	$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}$
Cosine	$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$
Tangent	$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b}$
Cosecant	$\csc(\theta) = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{c}{a}$
Secant	$\sec(\theta) = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{c}{b}$
Cotangent	$\cot(\theta) = \frac{\text{adjacent}}{\text{opposite}} = \frac{b}{a}$

where a is the opposite side, b is the adjacent side and c is the hypotenuse

Common Trigonometric Values

Memory trick: write $\sin$ as $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$ for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$. Then $\cos$ is the **same list reversed**, and $\tan = \frac{\sin}{\cos}$.

Angle	$\sin$	$\cos$	$\tan$
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	undefined

- $\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \approx 0.707$ and $\frac{\sqrt{3}}{2} \approx 0.866$.
- $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \approx 0.577$ and $\sqrt{3} \approx 1.732$.
- $\tan(90^\circ)$ is **undefined** because $\cos(90^\circ) = 0$ (you cannot divide by zero).

Pythagorean Triples (worth memorising)

Whole-number side lengths that satisfy $a^2 + b^2 = c^2$ — spotting them saves calculation: - 3, 4, 5 and its multiples (6, 8, 10 ; 9, 12, 15 ; 15, 20, 25) - 5, 12, 13 - 8, 15, 17 - 7, 24, 25

Special Right Triangles

Triangle	Angles	Side ratio (opposite each angle)
Isosceles right	$45^\circ : 45^\circ : 90^\circ$	$1 : 1 : \sqrt{2}$
Half-equilateral	$30^\circ : 60^\circ : 90^\circ$	$1 : \sqrt{3} : 2$

4. Counting & Probability

Permutation and Combination Formulas

Description	Formula
Factorial	$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1$
Permutation of k objects from n distinct objects (order matters)	$nPk = \frac{n!}{(n-k)!}$
Combination of k objects from n distinct objects (order does not matter)	$nCk = \frac{n!}{k! \cdot (n-k)!}$
Permutation with repetition allowed (length k from n choices)	n^k
Circular arrangement of n distinct objects (clockwise/anticlockwise considered different)	$(n-1)!$
Circular arrangement where mirror images are considered same (necklace type, $n > 2$)	$\frac{(n-1)!}{2}$

Notes

- Use nPk when position/order is important.
- Use nCk when only selection matters, not order.
- For circular arrangements around a table, one fixed reference position removes rotational duplicates, so use $(n-1)!$.
- Handshakes** in a group of n people $= \frac{n \cdot (n-1)}{2}$ (this is just $nC2$ — each pair shakes once).
- Diagonals** in a polygon with n sides $= \frac{n \cdot (n-3)}{2}$.

Probability

Description	Formula
Probability of an event	$\frac{\text{favourable outcomes}}{\text{total outcomes}}$

- Probability is always between 0 (impossible) and 1 (certain).
- $P(\text{not } A) = 1 - P(A)$.

5. Reasoning Aids (Verbal & Numerical)

Letter–Number Values ($A = 1 \dots Z = 26$)

Used constantly in **numerical and verbal reasoning** codes (e.g. shift each letter forward by 3).

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

- **Anchor letters to recall fast:** $E = 5$, $J = 10$, $O = 15$, $T = 20$, $Y = 25$, and $Z = 26$.
- The vowels are $A = 1$, $E = 5$, $I = 9$, $O = 15$, $U = 21$.
- **Reverse value** ($Z=1 \dots A=26$): reverse value = $27 - (\text{normal value})$. e.g. $C = 3$, reverse = 24 .

Roman Numerals

Symbol	I	V	X	L	C	D	M
Value	1	5	10	50	100	500	1000

How to read: add symbols left to right; but when a smaller symbol comes **before** a larger one, you **subtract** it.

- Subtractive pairs: $IV = 4$, $IX = 9$, $XL = 40$, $XC = 90$, $CD = 400$, $CM = 900$.
- A symbol repeats at most **three** times in a row ($III = 3$, but 4 is IV , not $IIII$).
- Examples: $XXVII = 27$, $XLIX = 49$, $MMXXIV = 2024$.
- Memory phrase for the order: **I**vy **V**ines **X**tend **L**ush **C**limbing **D**own **M**aples.